

Objectives

- Explain how to resolve the conflict during merge.

In this hands-on lab, you will learn how to:

- Implement conflict resolution when multiple users are updating the trunk (or master) in such a way that it results into a conflict with the branch's modification.

Prerequisites

The following are the pre-requisites to complete this hands-on lab:

- Hands-on ID: **"Git-T03-HOL_001"**

Notes*:

Please follow the below steps for creating a free account in GitHub.
Do not use cognizant credentials to login to GitHub.

Estimated time to complete this lab: **30 minutes**.

Please follow the instructions to complete the hands-on. Each instruction expect a command for the Git Bash.

1. Verify if master is in clean state.
2. Create a branch **"GitWork"**. Add a file "hello.xml".
3. Update the content of "hello.xml" and observe the status
4. Commit the changes to reflect in the branch
5. Switch to master.
6. Add a file **"hello.xml"** to the master and add some different content than previous.
7. Commit the changes to the master
8. Observe the log by executing **"git log --oneline --graph --decorate --all"**
9. Check the differences with Git diff tool
10. For better visualization, use P4Merge tool to list out all the differences between master and branch
11. Merge the bran to the master
12. Observe the git mark up.
13. Use 3-way merge tool to resolve the conflict
14. Commit the changes to the master, once done with conflict
15. Observe the git status and add backup file to the .gitignore file.
16. Commit the changes to the .gitignore
17. List out all the available branches
18. Delete the branch, which merge to master.
19. Observe the log by executing **"git log --oneline --graph --decorate"**

SOLUTION

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SQL HISTORY TASK MONITOR
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT> cd WEEK-6
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6> cd 04.GIT
```

```
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git init
Initialized empty Git repository in C:/Users/valla/OneDrive/Desktop/DN-JAVA-FSE-REACT/WEEK-6/04.GIT/.git/
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

```
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> echo "<hello>Initial Version</hello>" > hello.xml
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
```

```
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git commit -m "Initial commit"
[master (root-commit) 6d56a22] Initial commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 hello.xml
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git branch GitWork
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git branch
  GitWork
* master
● PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git checkout GitWork
Switched to branch 'GitWork'
```

```
hello.xml x
WEEK-6 > 04.GIT > hello.xml
1 <hello>Initial Version</hello>
2 |
```

```
hello.xml X
WEEK-6 > 04.GIT > hello.xml
1 <hello>Master Modified Version</hello>
```

```
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git add .
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git commit -m "Updated hello.xml in GitWork"
[GitWork e6c2e9c] Updated hello.xml in GitWork
1 file changed, 0 insertions(+), 0 deletions(-)
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git checkout master
Switched to branch 'master'
```

```
hello.xml X
WEEK-6 > 04.GIT > hello.xml
1 <hello>Master Modified Version</hello>
2 <hello>Branch Version</hello>
```

```
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git commit -m "Updated hello.xml in master"
[master 3d3ae85] Updated hello.xml in master
1 file changed, 0 insertions(+), 0 deletions(-)
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git merge GitWork
warning: Cannot merge binary files: hello.xml (HEAD vs. GitWork)
Auto-merging hello.xml
CONFLICT (content): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git add hello.xml
• PS C:\Users\valla\OneDrive\Desktop\DN-JAVA-FSE-REACT\WEEK-6\04.GIT> git commit -m "Resolved merge conflict"
[master 4f9ff9f] Resolved merge conflict
```